EE6221 CA2 Assignment

State of the Art of General-Purpose Humanoid Robots:
Robot Brains and the Limits of Real-World Deployment

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Abstract

This report reviews the current state of general-purpose humanoid robots, with a focus on one technology that decides whether they are useful: the control stack that turns camera images and a spoken instruction into coordinated motion of every joint. I compare commercial platforms (Boston Dynamics Atlas, Figure 03, Apptronik Apollo, Unitree G1) and recent research on learned whole-body control and vision-language-action (VLA) models (GR00T N1, π_0, Gemini Robotics 1.5, Helix 02, FAST, HuGe), as well as recent tests of general-purpose frontier models, mainly OpenAI's GPT-6 Astra but also Claude, Gemini and Grok, used as a robot brain. I also compare where each brain can run (on the robot or in the cloud) and how fast it can react, since many applications require offline, on-device operation within a fixed latency budget. I argue that the best current solution is a *hierarchical, learned* architecture in which a slow reasoning model, a fast visuomotor policy, and a very fast whole-body controller are trained to work together, as exemplified by Figure's Helix 02, while Atlas is the most deployment-ready hardware. I then discuss why humanoids are still mostly at the pilot stage while AI agents are already used widely in software, the main open challenges (data, reliability, safety certification, energy, and evaluation), likely future directions, and what I learned.

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1 Introduction and scope

A humanoid robot has a human-like body: two legs, a torso, two arms and, in most current designs, multi-fingered hands. The case for this form is practical rather than aesthetic. Factories, warehouses and homes were built for human bodies, so a robot with the same reach, height and hand shape can in principle use existing tools, shelves and doorways without redesigning the workplace. The same argument explains why humanoids are hard: a body with 50 or more actuated joints that must balance on two small feet is underactuated, has highly coupled dynamics, and must keep doing so while the hands interact with unknown objects.

For many years the hardware was the bottleneck. Today several companies can build reliable humanoid bodies, and the limiting factor has moved to the "brain": how the robot perceives a scene, decides what to do, and converts that decision into stable whole-body motion. This review therefore treats the overall humanoid solution but concentrates on that control stack. It connects directly to EE6221: forward and inverse kinematics of redundant arms, robot dynamics and control, vision sensors and pose estimation, sensor fusion with the Kalman filter, and vision-based control all appear as layers inside a modern humanoid.

Method. I compared systems using public technical reports, peer-reviewed or preprint papers, and official company announcements. Company announcements are treated as *claims*; I note where results lack independent verification. Each solution is judged on five criteria: (i) breadth of tasks it can perform without task-specific programming, (ii) length and reliability of autonomous operation, (iii) data efficiency and ability to transfer across robots, (iv) readiness for real deployment (hardware maturity, safety, integration), and (v) openness and quality of the evidence.

2 Background: the layers of a humanoid control stack

It is useful to describe any humanoid as a stack of loops running at very different rates.

- **L1 Actuation and state estimation** (about 1 kHz). Joint torque or current control, plus estimation of the floating base pose from IMU, joint encoders and foot contact, typically with a Kalman-type filter. The base pose is not measured directly, so this estimate is what every higher layer trusts.
- **L2 Whole-body control (WBC)** (about 50–500 Hz). Given a desired motion of the hands, feet or centre of mass, compute joint commands that achieve it while keeping balance and respecting joint and contact limits.
- **L3 Visuomotor policy** (about 10–200 Hz). Map camera images and proprioception to short-horizon targets such as hand poses, grasp commands, or a reference motion for the WBC.
- **L4 Task reasoning** (about 1–10 Hz or slower). Interpret a language instruction, break it into steps, track progress, and recover from failure.

Classical humanoids built L2 from models: zero-moment-point (ZMP) walking, model predictive control (MPC) of a simplified centroidal model, and a quadratic-program WBC that resolves the redundancy of the kinematic chain (the same null-space idea used for redundant manipulators, but with contact and balance constraints added). L3 and L4 were hand-written state machines. The common thread of the last two years is that *every layer above L1 is being replaced or augmented by learned models*, and the key design question is how those learned layers are divided and connected.

3 Review of the state of the art

3.1 Hardware platforms

Boston Dynamics Atlas (product version, January 2026) is the most deployment-ready body: 56 degrees of freedom, 50 kg payload, water resistance, operation from −20 to 40 °C, autonomous battery swapping, and connection to factory software through the Orbit fleet platform. All 2026 production goes to Hyundai and Google DeepMind [1]. **Figure 03** (October 2025) was redesigned for volume manufacturing and homes, with palm cameras and tactile hands, and is the body for Helix 02 [23]. **Apptronik Apollo** is one of the robots used by Gemini Robotics 1.5 [20], and the low-cost **Unitree G1** has become the common research platform [5, 19].

3.2 Learned whole-body control

Reinforcement learning in GPU-parallel simulation, followed by sim-to-real transfer, is now the standard way to obtain robust humanoid locomotion. The newest trend is the *behaviour foundation model*, a whole-body controller pre-trained on large motion datasets so it can be reused across tasks [4]. FAST [3] adds a small, constrained residual policy for new motions and centre-of-mass objectives, which shows that learned controllers still benefit from classical balance quantities.

3.3 Vision-language-action models

A VLA model starts from a vision-language model pre-trained on internet data and is trained further to output robot actions.

- π_0 (Physical Intelligence, 2024) [22]: a vision-language backbone with a flow-matching action expert, trained across many robot types; it set the recipe later models follow.
- **GR00T N1** (NVIDIA, 2025) [21]: an open-weight humanoid model with a dual-system design, a vision-language "System 2" and a diffusion action "System 1".
- **Gemini Robotics 1.5** (Google DeepMind, 2025) [20]: a VLA plus a separate embodied-reasoning planner, with motion transfer that runs the same skill on different robots, including Apollo.
- **Helix 02** (Figure, 2026) [2]: adds a learned whole-body controller ("System 0") beneath Helix's System 1 and System 2. Figure showed a four-minute autonomous dishwasher task across a kitchen and later claimed multi-hour shifts; these are company claims without independent evaluation.
- **HuGe** (October 2026) [5]: pre-trains humanoid VLAs on over 2,000 h of egocentric human and robot video, outputting motion tokens for a pre-trained whole-body controller; on a real G1 the human data improved pick-and-drop success.

3.4 General-purpose frontier models as the robot brain

A newer idea, which appeared in late 2026, skips robot-specific training entirely. Instead, a general-purpose frontier model, the same kind of model behind today's chatbots and AI agents, is connected to a robot and asked to control it. The most tested example is OpenAI's GPT-6 Astra, which several research groups studied within a few weeks of each other:

- **HomeBody** (Stanford and Caltech, September 2026) [16] runs a Unitree G1 humanoid with Astra. Astra does not output joint commands. It picks a skill (grasp, navigate, open a drawer) and its target through a structured tool call, and the skills run on a local laptop

GPU. The robot tidied an unfamiliar kitchen, but the authors note pauses between skills caused by Astra's reasoning latency, extra API cost, and overheating finger servos. It is a project page with code, not a paper with measured results.

- **Astranomy** (NVIDIA, October 2026) [17] tests Astra as a robot policy in simulation without any robot training. Used as an agent with tools, it reached a 74.8% success rate on the RoboLab-120 benchmark, against 28.0% for π_0's successor $\pi_{0.5}$. The cost is speed and money: a single direct call takes about 18.6 s on average, and the agent version uses about 30.9 million tokens (about US$36) per episode.
- **SenseNova's joint study** (October 2026) [18] evaluates Astra on 8 robot configurations and 12 benchmarks, including humanoids. It finds that the best setup uses Astra as a slow "System 2" planner coordinating a fast "System 1" controller such as $\pi_{0.5}$, which it calls "costly but effective". The authors conclude that Astra is not yet practical embodied intelligence, because of cost, execution time and weak recovery from irreversible failures.

These results show that a general model can understand scenes and plan tasks surprisingly well with no robot training. But it is far too slow to close a balance or grasping loop. In every humanoid example it sits at the top of the stack and calls faster controllers below it. All three sources are preprints or project pages, and I did not find an official technical report from OpenAI describing Astra's design.

Other frontier models. Astra is not the only general model being tried as a robot brain.

- **Gemini.** Google builds its own robotics models on top of Gemini. Gemini Robotics-ER 1.5 is a Gemini model tuned for embodied reasoning and used as the planner above the Gemini Robotics VLA (Section 3.3) [20].
- **Claude.** Robocurve compared Claude Fable 5, Fable 5.1 and Astra on the same pair of 6-DoF YAM arms, with the model sending end-effector poses directly [9]. On placing a block in a bowl (20 trials each), Astra succeeded 19 times, Fable 5.1 8 times and Fable 5 once. On a harder puzzle task, Astra and Fable 5.1 each succeeded only 2 times in 20. The authors note the runs were not interleaved, used different rigs, and were graded by an operator, so this is a small, early comparison.
- **Grok.** Tesla plans to use xAI's Grok as the conversational "brain" of its Optimus humanoid, while a vision network adapted from Tesla's self-driving software handles perception and movement [11, 12]. Musk has said that for a Grok-level conversation Optimus needs a Wi-Fi or cellular connection to the cloud [13], which is exactly the on-device versus cloud trade-off discussed in Section 4. This is a plan, not a published or tested system.

The pattern is the same for every company: the general model supplies language understanding and planning, and something faster and more specialised moves the body.

3.5 Comparison

Table 1 summarises the solutions against the criteria in the introduction.

4 Deployment constraints: where the brain runs and how fast it must react

Section 3 compared the solutions by what they can do. Before using one, two practical questions also matter. First, what is the *latency budget*, the longest allowed delay between a change in the environment and the robot's response? Second, must the brain run *on-device*, on the robot's own

Solution	Architecture	Main strength	Evidence
Atlas (product)	Industrial humanoid; DeepMind models being integrated	Deployment readiness: payload, environment rating, battery self-swap, fleet software	Company specs; first fleets in 2026
GR00T N1	Dual-system VLA (VLM + diffusion action head)	Open weights; real + human-video + synthetic data	Paper; sim and real-robot tests
Gemini Robotics 1.5	VLA + separate embodied-reasoning model	Reasoning before acting; cross-robot motion transfer	Technical report
Helix 02 + Figure 03	Three tiers: S2 reasoning, S1 visuomotor, S0 learned WBC	Single learned system from pixels to whole-body action; long-horizon demo	Company demos only
FAST	Pre-trained WBC + constrained residual adaptation	Fast adaptation without forgetting; CoM-aware balance	Preprint; sim and real
HuGe	VLA → latent motion tokens → WBC	Scales data with egocentric human video	Preprint; benchmark + G1
General frontier models (GPT-6 Astra, Claude Fable, Gemini-ER, Grok)	General model as planner, calling skills or a fast policy	No robot training needed; strong task planning; but slow and costly	Preprints, project pages, small lab tests; Grok only announced

Table 1: Comparison of the reviewed humanoid solutions.

computer, and must it work *offline*, with no network connection? The answers depend on the application, and they decide which models can be used at all.

Situation	Example	Latency budget	Where the brain can run
Balance, fall recovery	Slipping or being pushed	about 1–10 ms	On-device only
Contact and grasp correction	An object slips in the hand	about 10–50 ms	On-device
Reacting to people	A person steps into the robot's path	about 100 ms	On-device
Task planning	"Tidy the kitchen"	about 1–10 s	Cloud is acceptable if a network is available
Offline sites	Factory with no internet or strict data privacy; disaster site	Whole stack	Everything on-device, so smaller models

Table 2: Indicative latency budgets by situation. These are my estimates, based on the loop rates in Section 2, not values from a single source.

Table 3 lists what each source states about model size, speed and where it runs. Many companies do not publish these numbers.

Three points follow from the two tables.

1. **Smaller networks run faster, and the pattern is very regular.** Helix 02's 10M-parameter System 0 runs at 1 kHz, Helix's 80M-parameter System 1 at 200 Hz, the 2–7B models at about 7–10 Hz, and the cloud frontier model Astra takes about 18.6 s per decision. Each step up in size buys more reasoning but costs about an order of magnitude in speed.

System	Parameters	Rate / delay	Where it runs	Training data
Helix [6]	S2: 7B; S1: 80M	S2: 7–9 Hz; S1: 200 Hz	Fully onboard, embedded low-power GPUs	About 500 h teleoperation
Helix 02 [2]	S0: 10M; others not disclosed	S1: 200 Hz; S0: 1 kHz	Not stated	Over 1,000 h human motion (for S0)
GR00T N1 [21]	2.2B (1.34B in the VLM)	VLM: 10 Hz; action head: 120 Hz; 63.9 ms per 16-action chunk	Timed on an L40 data-centre GPU	Real, human-video and synthetic
π_0 [22]	3.3B	Up to 50 Hz	Mobile robot: off-board over Wi-Fi	About 10,000 h demonstrations
Gemini Robotics On-Device [7]	Not disclosed	"Low-latency", no figure	On the robot; works with no network	Adapts from 50–100 demos
HuGe [5]	Qwen-based VLM; total not stated	50 Hz action timeline	Not stated	Over 2,000 h egocentric video
GPT-6 Astra [17]	Not disclosed	About 18.6 s per call	Cloud API	No robot-specific training

Table 3: Size, speed and location of the reviewed robot brains, as stated in each source.

2. **The latency budget decides which layer a model can be.** Anything that must react within about 100 ms has to be a small network on the robot. Only task planning can tolerate cloud delay, which is why every Astra system in Section 3.4 used Astra as the planner only.
3. **An offline requirement rules out cloud models completely.** Then the top layer must be a model of a few billion parameters that fits on the robot's computer, as in Helix's onboard 7B System 2 or Gemini Robotics On-Device, which is designed to work with intermittent or zero connectivity. For scale, NVIDIA's Jetson AGX Thor, a computer built for humanoids, gives up to 2,070 TFLOPS (sparse FP4) within a 40–130 W power budget and 128 GB of memory [8]. That is enough for several-billion-parameter models, but it uses battery power that would otherwise move the robot.

This gives a second reason for the hierarchical stack. Because the layers are separate, the fast layers can stay on the robot while the slow planner runs in the cloud when a network is available. I suggest that a practical design should also carry a smaller onboard planner as a fallback, so the robot keeps working, more simply, when the connection drops.

5 The best solution and why

In my view the best current solution is the **hierarchical learned stack**: a slow reasoning model (System 2), a fast visuomotor policy (System 1) and a learned whole-body controller (System 0), trained so that the layers share a common interface. The most complete published example is **Figure's Helix 02 on Figure 03**. For a factory buyer today, **Atlas** is the most deployment-ready body, and its pairing with DeepMind's models points to the same architecture. I choose this solution for four reasons.

1. **It matches the physics of the problem.** Balance must be corrected within milliseconds,

grasping within tens of milliseconds, and task planning within seconds. One network running at one rate cannot serve all three. The three-tier design gives each timescale its own loop. This is the same principle as the cascaded inner and outer loops we study in control, applied to learned components. Section 4 shows that the same split also decides where each layer can run: the fast loops on the robot, and the slow planner on the robot or in the cloud.

2. **It joins walking and manipulation.** Older humanoids switched between separate locomotion and arm controllers, so they walked, stopped and only then reached. A learned System 0 lets the robot walk while carrying and balance during contact; the dishwasher demonstration is the strongest public evidence of this.

3. **It scales with data, not engineering.** A new task needs demonstrations rather than a new controller, and GR00T N1 and HuGe show that human video and synthetic data can supplement scarce robot data.

4. **Independent groups converged on it.** GR00T N1, Gemini Robotics 1.5, HuGe and even the GPT-6 Astra studies (Section 3.4) all place a slow planner above faster controllers.

I rank Helix 02 first as an integrated system but with a clear qualification: its evidence is company demonstrations, with no public benchmark or paper. If openness and reproducibility are weighted more heavily, GR00T N1 (open weights) and Gemini Robotics 1.5 (detailed technical report) are better references for how the architecture works.

6 Challenges in improving humanoid robots

Why are there so few real applications? LLMs and AI agents are already part of everyday software work, such as writing code and answering customers. Humanoids, which use the same kind of models as their "brain", are still mostly at the pilot stage. The public record is small. Agility Robotics and the logistics company GXO signed what they called the first humanoid Robot-as-a-Service contract in 2024 [15]. Figure reports about 740 robots operating by the end of June 2026, which is a company figure [14]. Tesla is still installing its first Optimus production lines, and analysts expect external sales no earlier than the second half of 2027 [12].

In my view the gap comes from four differences between software agents and robots:

- **Mistakes cost more.** A wrong line of code can be reviewed, tested and undone. A wrong physical action can break equipment or hurt someone, and it cannot be undone.
- **Each copy costs money.** A software agent can be copied to a million users almost for free. Every robot is hardware that must be built, charged, maintained and insured.
- **The bar is a working alternative.** A humanoid must beat a fixed automation cell or a human worker on total cost and reliability, and both already work well.
- **There is less data and less certainty.** Robot data is scarce, and there is no safety standard yet for certifying the robot.

The challenges below are the technical side of this gap. Note that LLM progress does reach robots quickly, but only at the top of the stack, as the planner. The body, the fast control loops and safety are what still limit commercial use.

1. **Data scarcity and embodiment gap.** Language models learned from trillions of words; humanoid policies have thousands of hours of demonstrations. Teleoperation is slow and expensive, and human video does not contain the robot's joint torques or contact forces. HuGe's need for simulation replay and arm re-rendering shows how much work is required just to make human data usable.

2. **Reliability in the long tail.** A demo succeeding once is very different from 99.9% success over an eight-hour shift. Learned policies can fail in surprising ways when lighting, objects or clutter differ from training data, and these failures are hard to predict in advance.
3. **Safety and certification.** A humanoid that falls when power is lost is a new hazard class. Existing standards do not cover it: ISO 10218 does not address mobility, ISO 3691-4 excludes these robots, and ISO 13482 is not for industrial sites. The dedicated standard for dynamically stable industrial mobile robots, ISO 25785-1, is still in development [24]. It is also unclear how to certify a neural network controller whose behaviour cannot be fully specified in advance. General models make this harder. In Robocurve's RoboHarm test (September 2026), Claude Fable 5.1, GPT-6 Astra and the robot model MolmoAct2 controlled real arms and were each given 5 instructions a safe robot should refuse, 20 times each [10]. Fable refused 20 of 100, all on one task (stabbing a doll); Astra refused 2; MolmoAct2 refused none. Fable and Astra each pushed a metal screwdriver into a toaster 6 and 7 times out of 20. The chatbot safety training of these models does not carry over to physical actions, so a separate safety layer on the robot is still needed.
4. **Hands and touch.** Dexterous hands are fragile and costly, and tactile sensing is only now entering learned policies.
5. **Energy and cost.** Onboard compute competes with the battery used for motion (Section 4), and unit cost must fall far enough to beat an automation cell or a human worker.
6. **Evaluation.** Most impressive results are company videos; without shared benchmarks and independent tests, systems are hard to compare.

7 Future directions

- **Data from humans and simulation.** Egocentric human video (as in HuGe), synthetic data and world models should reduce dependence on slow teleoperation.
- **Learning from deployment.** Fleets connected through software such as Orbit can pool corrections, so one robot's lesson improves all of them.
- **Learning plus model-based guarantees.** Learned policies wrapped by verifiable safety filters (balance or collision constraints, as FAST's centre-of-mass objectives hint) are the likely route to certification.
- **Standards and benchmarks.** Completing ISO 25785-1 and shared benchmarks such as HumanoidArena will move the field from demos to measurable progress.
- **Efficient on-board models.** Distilled VLAs and dedicated accelerators will let more of the stack run offline within the latency budgets of Section 4.

8 What I learned from this review

- **The course material is still there, just in a new place.** Before this review I expected AI to have replaced classical robotics. It has not. The kinematics we did in class (frames, D–H parameters, forward kinematics) is what retargeting uses to map a human motion onto the robot's joints. Dynamics and balance quantities such as the centre of mass are fed into the learned whole-body controller. Estimating where the robot's floating base is and how fast it moves is still a filtering problem, like the Kalman filter in Part 2.
- **Timescale is the key design choice.** Every strong system I read about splits the work into a slow planner, a middle layer and a fast controller. This is the cascaded-loop idea from control, only now each loop is a neural network.

- **Data is the real bottleneck.** Many recent papers look like algorithm papers, but what they really do is find new data: human video, simulation, or a cheaper way to teleoperate.
- **Not all sources are equal.** I learned to check what kind of evidence I was reading. A peer-reviewed paper, a preprint, a project page and a company video support a claim to very different degrees, and most humanoid "results" are in the last two groups.
- **What surprised me most.** First, GPT-6 Astra beat a model trained on robot data (74.8% against 28.0% success in simulation) without any robot training at all, yet one call took about 18.6 s. The smartest part of the system is also the slowest, so it can only work if fast controllers sit below it. Second, in the RoboHarm test the general models refused very few unsafe physical commands, even though they refuse harmful requests in chat. I had assumed their safety training would carry over to the robot, and it does not.

9 Effect of this assignment on my understanding and interest

[This section must be your own honest answer; the points below are prompts only.]

- **Understanding of robotics and intelligent systems:** [Did it increase? Which link between the lectures (kinematics, dynamics and control, vision sensors, Kalman filtering, vision-based control) and real humanoids became clearer?]
- **Interest in a career in robotics and intelligent systems:** [Did it increase? Which part of the stack (control, perception, learning, safety) would you want to work on, and why?]

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